
ANOMALY DETECTION IN NETWORK TRAFFIC USING MACHINE LEARNING**Chaitanya Santosh Kumbhar^{*1}, Sakshi Anandroa Kashid^{*2}**^{*1,2}Student, ASM's CSIT College, Chinchwad Pune, India.

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ABSTRACT

As internet usage and digital infrastructures have proliferated, cyber threats have become increasingly sophisticated and pervasive. Conventional intrusion detection systems are heavily signature-based, which is extremely limited in the coverage of unknown or zero-day attacks. Here we present an unsupervised machine learning-based anomaly detection system which can detect anomalous patterns in network traffic data. One of the main models which we are using in this study is Isolation Forest which is an algorithm that detects anomalies by isolating abnormal data points from the normal data points contained in network traffic datasets. The detection capability can further be enhanced using an Autoencoder-based deep learning model. To find departures from typical network behavior, the system examines network characteristics like protocol type, packet size, connection duration, and traffic flow statistics. In real-world network environments, the suggested framework makes threat detection automated and scalable. When compared to conventional intrusion detection techniques, experimental evaluation shows that machine learning-based anomaly detection systems can greatly increase detection accuracy and decrease false positives.

I. INTRODUCTION

Contemporary computer networks are essential for personal communication, government infrastructure, and business operations. However, the growing reliance on digital networks has also led to a notable increase in cyberthreats like malware infections, DDoS attacks, and unauthorized intrusions. Because they rely on predetermined attack patterns, traditional network security mechanisms like firewalls and signature-based intrusion detection systems (IDS) have limitations. New or unseen attacks cannot be successfully detected by these systems. An intelligent method for identifying unusual patterns in network traffic is machine learning. Machine learning models identify deviations from typical behavior by identifying patterns in network data rather than depending on pre-established rules. This study focuses on identifying anomalies in network traffic using unsupervised machine learning techniques. The system's goal is to help network administrators stop possible cyberattacks by automatically identifying suspicious activity.

II. METHODOLOGY

Data collection, data preprocessing, feature extraction, model training, and anomaly detection are some of the steps in the proposed research's methodology.

Data Collection

Gathering information about network traffic is the first stage in the anomaly detection process. In this study, the model is trained and assessed using publicly accessible cybersecurity datasets. Typical datasets consist of:

- **KDD Cup 99 Dataset**
- **NSL-KDD Dataset**
- **CICIDS2017 Dataset**

Network traffic records from both typical and malicious network activity, including intrusion attempts, denial-of-service attacks, and unauthorized access, are included in these datasets.

Data Preprocessing

Inconsistent formats, missing values, and redundant information are common in raw network traffic data. Preprocessing is therefore necessary prior to feeding the data into machine learning models. Included in the preprocessing phase are:

- Data cleaning and removal of missing values
- Feature normalization and scaling
- Conversion of categorical features into numerical values
- Removal of irrelevant or redundant features

Data preprocessing raises the dataset's quality and boosts machine learning model performance.

Feature Selection

Because it aids in determining the most pertinent characteristics for anomaly detection, feature selection is a crucial stage in machine learning. Various aspects of network traffic behavior are represented by the chosen features. Among the crucial elements employed in this study are:

- Protocol type (TCP, UDP, ICMP)
- Packet size
- Connection duration
- Source IP address
- Destination IP address
- Number of packets transmitted

These features help the model distinguish between normal and abnormal network activities.

III. MODELING AND ANALYSIS

The anomaly detection system that the research proposed primarily relies on the Isolation Forest algorithm. Isolation Forest is an unsupervised machine learning algorithm that specifically focuses on anomaly detection. Unlike other machine learning algorithms for classification tasks, the Isolation Forest algorithm does not need any training data. This algorithm works by randomly partitioning the dataset with the help of a decision tree. Normal data points need to be partitioned more to be isolated, while anomalous data points are easily isolated since they deviate from the majority of the data. An anomaly score is given to the data point based on how easily it can be isolated. In addition to the Isolation Forest algorithm, the system can also be integrated with an Autoencoder neural network. Autoencoder is a type of deep learning algorithm that learns the compressed version of the input data. The network learns to reconstruct the normal traffic pattern. Once anomalous traffic is detected, the error level increases, indicating the presence of anomalous traffic. The Isolation Forest algorithm and the Autoencoder algorithm together form a strong system for detecting abnormal network traffic patterns.

System Architecture:

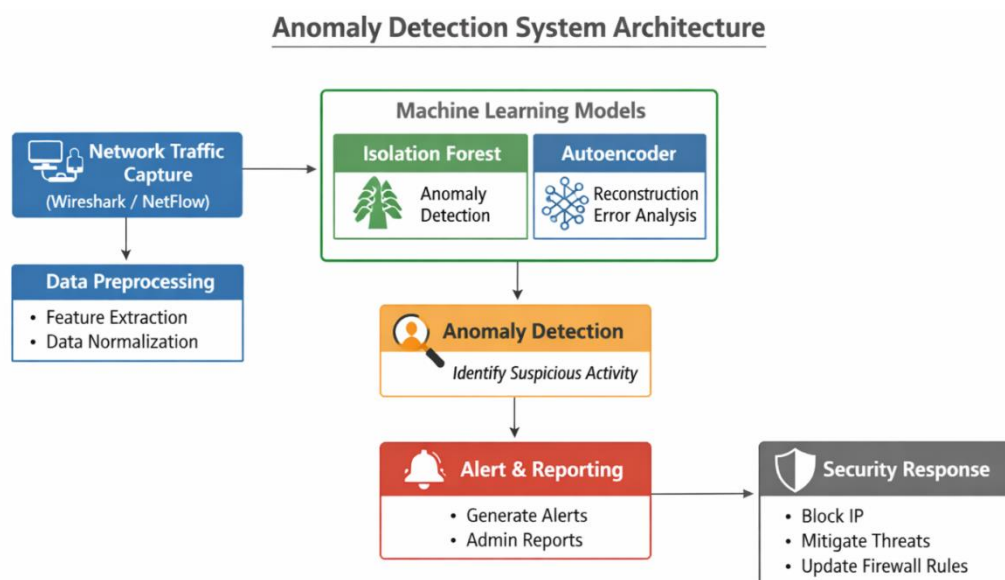


Figure 1: Anomaly detection system Architecture

The architecture of the proposed anomaly detection system consists of the following components:

- Network traffic capture module
- Data preprocessing module
- Feature extraction module
- Machine learning model
- Anomaly detection and alert system

Figure 1 illustrates the overall architecture of the proposed anomaly detection system.

IV. RESULTS AND DISCUSSION

It is worth noting that the proposed anomaly detection model was evaluated using network traffic data sets containing normal and malicious samples. Isolation Forest algorithm effectiveness in detecting abnormal patterns in data sets was demonstrated. The proposed model was successful in detecting abnormal network traffic patterns by assigning anomaly scores to network data points. Data points with high anomaly scores were identified as abnormal data points.

Evaluation of proposed model performance was done using various performance metrics commonly applied in machine learning models:

- Accuracy
- Precision
- Recall
- F1-score

The results obtained from the experiment show that the machine learning-based anomaly detection system performs better compared to the rule-based intrusion detection systems. The system is efficient in analyzing the large volume of network traffic data.

Figure 2, 3 & 4 shows the graphical representation of anomaly detection performance based on models

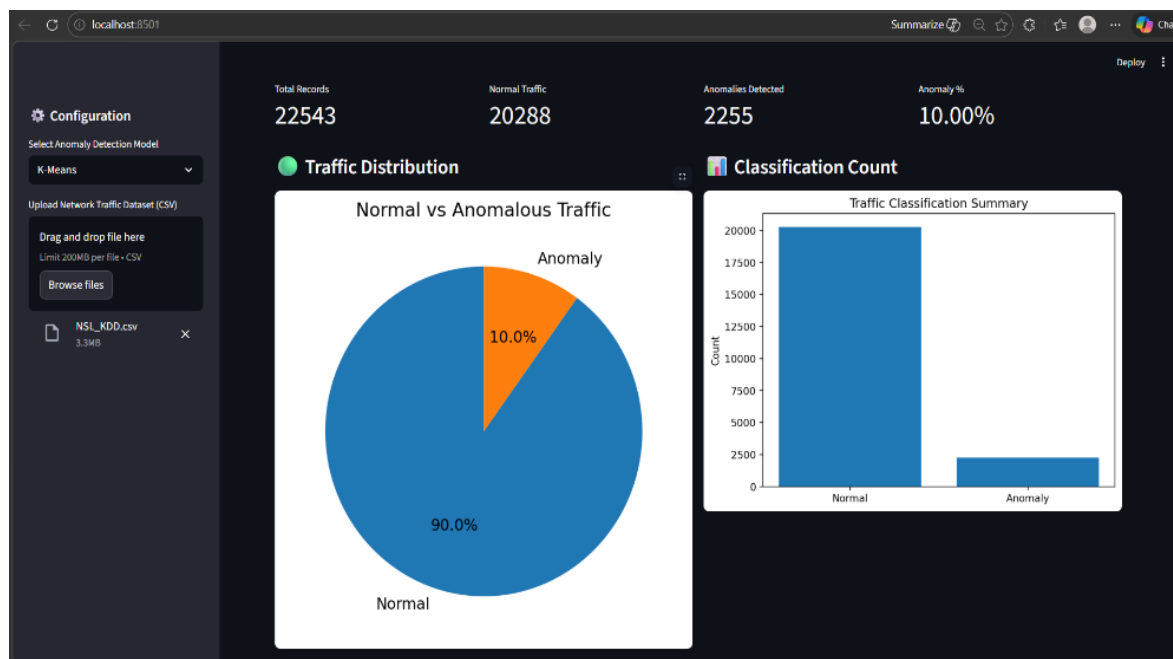


Figure 2: User interface

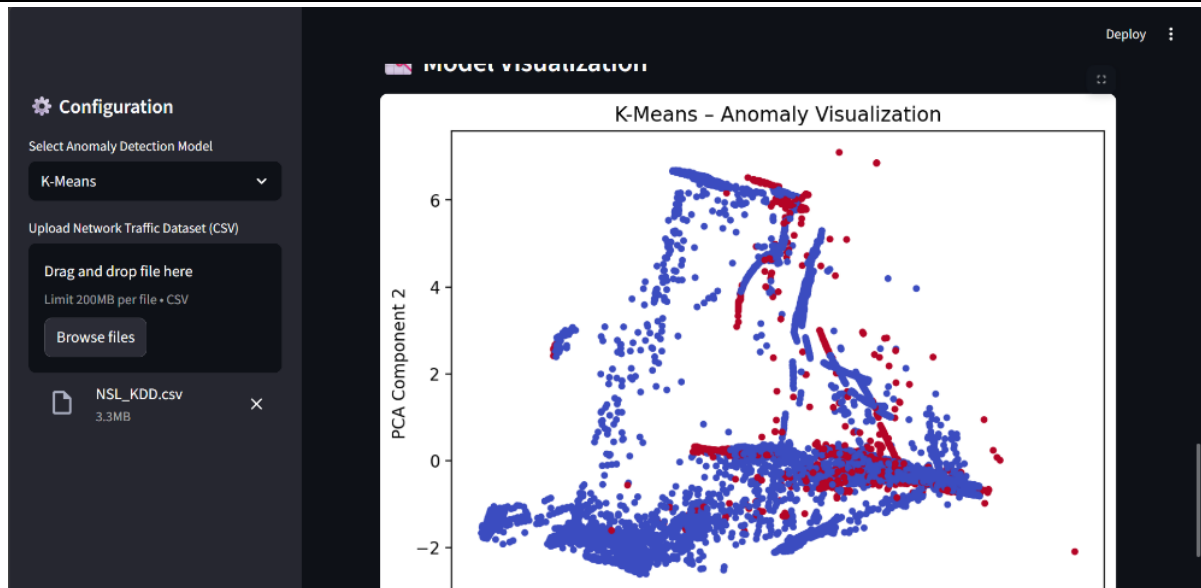


Figure 3: K-means for Anomaly visualization

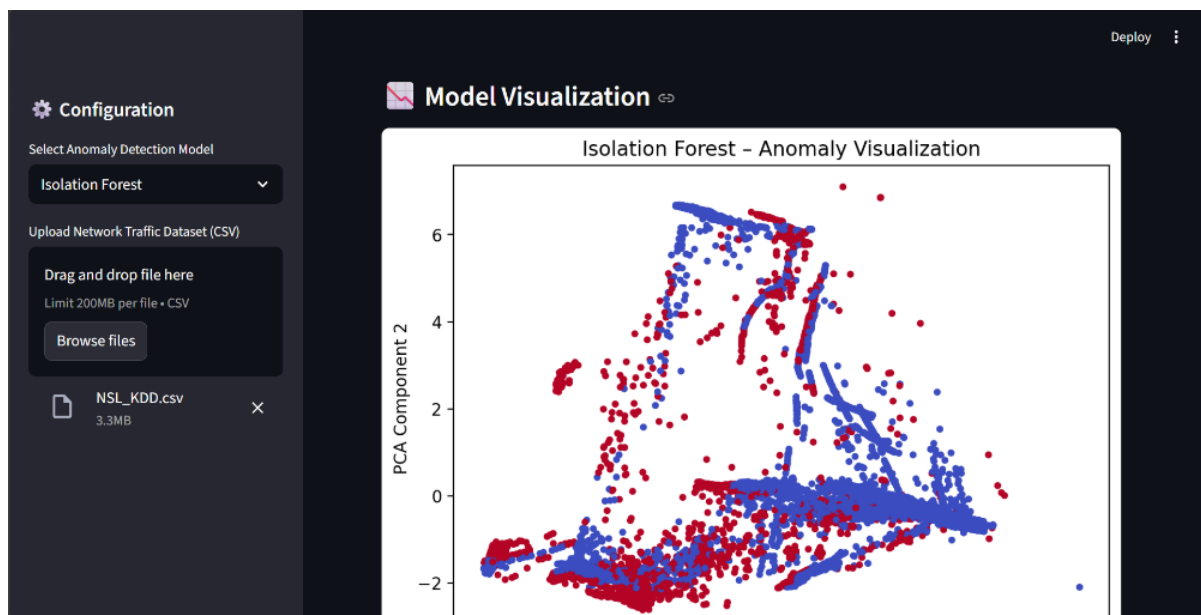


Figure 4: Isolation Forest for Anomaly visualization

V. CONCLUSION

This research proposed a machine learning-based anomaly detection system for detecting abnormal patterns in network traffic. Traditional intrusion detection systems use known attack signatures to perform detection; however, this makes it difficult to detect unknown threats in a network. The proposed system uses unsupervised machine learning techniques, such as Isolation Forest, to detect anomalies in network traffic. Experimental results show that a machine learning-based intrusion detection system is efficient in detecting abnormal network traffic patterns and thus enhances network security. The proposed system can be applied in a real-world scenario to improve network security against cyber attacks. Future research should be done on using deep learning techniques in conjunction with monitoring systems to improve the performance of anomaly detection systems.

VI. REFERENCES

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